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Transactions in SQL Server

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Working with Transactions in SQL Server

What is a Transaction?

A **transaction** is a group of operations treated as a single unit.
It **ensures data integrity** using **ACID** properties (Atomicity, Consistency, Isolation, Durability).

Real-Time Case Study: E-Commerce Checkout

Scenario: A customer buys items from an e-commerce store.

Transaction includes:

- Deduct stock from Products table.
- Create a new order in Orders.
- Deduct amount from UserWallet.

ACID Application:

- **Atomicity**: All steps succeed or none.
- **Consistency**: Inventory and payment constraints must be respected.
- **Isolation**: Another user can’t order the same product while it's being processed.
- **Durability**: Once the order is confirmed, it survives power failure.

Using COMMIT and ROLLBACK

Example:

```
BEGIN TRANSACTION;

BEGIN TRY
    UPDATE Accounts SET Balance = Balance - 100 WHERE AccountID = 1;
    UPDATE Accounts SET Balance = Balance + 100 WHERE AccountID = 2;

    COMMIT TRANSACTION;
END TRY
BEGIN CATCH
    ROLLBACK TRANSACTION;
    PRINT ERROR_MESSAGE();
END CATCH
```

Real-Time Case Study: Bank Fund Transfer

Scenario: Transfer ₹500 from Account A to B. Both updates must succeed, or none.

Example:

```
BEGIN TRANSACTION;

BEGIN TRY
    UPDATE Accounts SET Balance = Balance - 500 WHERE AccountID = 101;
    UPDATE Accounts SET Balance = Balance + 500 WHERE AccountID = 102;

    COMMIT;
    PRINT 'Transaction Successful';
END TRY
BEGIN CATCH
    ROLLBACK;
    PRINT 'Transaction Failed: ' + ERROR_MESSAGE();
END CATCH
```

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Coding Standards and Naming conventions in SQL Server

1. Capitalize SQL Keywords
2. Avoid using * in SELECT command , specify required columns in to improve performance and clarity.
3. Use meaningful aliases: Employee e, Department d etc…
4. Avoid reserved words(key words) as identifiers (table names, column names etc..)
5. Tables Names : Employees, Students, Reports
6. Use singular column names (FirstName, LastName)
7. Use -- for single-line and /* */ for block comments.
8. Use Proper Prefix : PK, FK, DEFLT, CHK (for constrains)
9. Use proper prefix for PL SQL items : trg (trigger), usp(userdefined stored procedure), udf (user defined function)
10. Boolean values column names / variable names should be prefixed with Is, Has, Can
11. Prefix parameters with @ (trg, usp, udf)

Q & A

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